Scaling and reading interpretable spline causal discovery: adaptive sparsity, parent screening, and shape-faithful edge functions

[authors to be confirmed]

Abstract

SPADE learns a non-linear additive-noise DAG whose edges are B-spline functions, so every learned dependency can be plotted and inspected. It has three practical weaknesses: its group-lasso weight must shrink with the number of variables d, it shrinks strong edges as much as weak ones, and its raw spline curves wiggle, which hides the shapes it is meant to expose. We address each. (i) An adaptive group-lasso averaged over half-sample fits (SPADE-AB) raises instantaneous-DAG AUROC by 0.014 on fresh seeds (Wilcoxon $p = 0.004$, $n = 45$ runs). (ii) A width-aware penalty $\lambda_g = \min(0.01, 0.2/d)$ with per-node parent screening keeps AUROC at 0.96 to 0.98 for $d = 50$ to 200, where the fixed penalty collapses to 0.58 to 0.82. (iii) A smoothness-penalized refit of the retained edges lifts the agreement between learned and true curve class from 0.12–0.25 to 0.93–0.94 without changing the recovered structure. On the Sachs protein-signalling network, the one real benchmark with a ground-truth graph that we use, SPADE-AB attains AUROC 0.664 against at most 0.566 for the other methods tested, although absolute accuracy remains low. We state the limits: all comparisons at $d \geq 100$ are against linear baselines (non-linear competitors do not run at that size), and the synthetic gains are modest.

1 Introduction

SPADE [Dang et al., 2026] couples Kolmogorov–Arnold-style B-spline edge functions [Liu et al., 2024] with an acyclicity constraint [Zheng et al., 2018] and a group-lasso, giving a non-linear causal graph whose edges are readable curves. Its published evaluation stopped at $d = 50$ and used one fixed sparsity weight. This paper asks three questions a practitioner would ask next. Does it work at hundreds of variables? Can the sparsity step be made less biased? And are the curves it draws actually faithful to the mechanisms that generated the data, which is the property that justifies calling the model interpretable? We answer with protocols fixed before the final runs, candidates chosen on validation data disjoint from the evaluation data, and failures reported.

2 Method

Base model. Each variable x_i is predicted only through spline edges of its candidate causes, $\hat{x}_i = b_i + \sum_{j \neq i} \phi_{ij}(x_j)$, trained with MSE, a group-lasso $\lambda_g \sum_{ij} \|\theta_{ij}\|_2$ over the coefficient vector of each edge, and the NOTEARS penalty on the edge-norm matrix [Zheng et al., 2018]. Edge scores are the standard deviation of each edge's contribution.

SPADE-AB (adaptive sparsity with averaging). Stage 1 is the base fit. Stage 2 refits with weights $w_{ij} = (\|\theta_{ij}\| + \epsilon)^{-1}$ normalized to unit mean (adaptive group-lasso, Zou 2006), which penalizes weak edges more and strong edges less. The final score is the mean of max-normalized scores over five fits on

Table 1: Fresh seeds (300–319 for $d = 6, 10$; 300–304 for $d = 20$): instantaneous-DAG recovery.

	d=6		d=10		d=20	
Method	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
SPADE	0.961	0.924	0.948	0.867	0.914	0.800
SPADE-AB	0.974	0.966	0.961	0.922	0.935	0.848

Table 2: Validation selection (seeds 200–219, $d = 6, 10$; 200–204, $d = 20$): mean AUROC.

Variant	d=6	d=10	d=20
base	0.958	0.923	0.871
adaptive	0.952	0.917	0.871
refit (no penalty)	0.930	0.900	0.855
bootstrap average	0.971	0.939	0.877
adaptive + average (SPADE-AB)	0.974	0.947	0.893

random half samples. On validation seeds neither the adaptive step nor the averaging alone was reliably better; the combination had the best mean AUROC at all three widths. Plain adaptive fitting was no better than the base fit and refitting without penalty was worse (Table 2).

Width-aware penalty and parent screening. The group-lasso sums over $d(d-1)$ edges, so a fixed λ_g over-penalizes as d grows. We use $\lambda_g(d) = \min(0.01, c/d)$, and before fitting keep for each node only the K candidate causes with the highest ExtraTrees importance, masking all other edges. $c = 0.2$ and $K = 15$ were chosen on validation graphs at $d = 50$ and 100.

Shape-faithful display refit. Raw spline coefficients are unconstrained at the grid tails and produce spurious wiggles. After structure is fixed, we refit the retained edges (top $2d$ by score) with a second-difference penalty $0.01 \sum \|D^2 \theta_{ij}\|^2$ and plot those curves. Structure (ranking) always comes from the unsmoothed fit, so the display step cannot change which edges are found.

3 Experiments

Instantaneous data come from a non-linear additive-noise generator with non-Gaussian noise (mechanisms $\tanh$, $\sin$, quadratic, rational). Validation and evaluation seeds are disjoint throughout. Baselines are run at their reference settings as in the original SPADE study.

3.1 Sparsity: SPADE-AB

Table 1 shows consistent gains. Pooled over the 45 runs the paired difference is +0.014 AUROC, with 24 wins, 13 ties and 8 losses over 45 runs (Wilcoxon $p = 0.0039$) . The gain per width is small and not individually significant; only the pooled test is. In an earlier comparison SPADE trailed NoGAM at $d = 10$ (0.926 vs 0.955); with SPADE-AB the gap on the same seeds narrows from 0.063 to 0.015 but does not close. *Environment note.* Re-running the base SPADE in a newer library environment gave $d = 10$ AUROC 0.892 on the official seeds, below the originally reported 0.926. Single-run variance across library versions is therefore about ± 0.03 at $d = 10$, comparable to some of the gains reported here, which is why we rely on paired, same-environment comparisons.

Table 3: Fresh seeds 600–602, $n = 3{,}000$. SPADE-scaled: $\lambda_g = \min(0.01, 0.2/d)$ and parent screening ($K = 15$). GOLEM and NOTEARS-MLP were run only at $d = 50$ (each fit exceeded 15 minutes); non-linear competitors (NoGAM, DAGMA-nonlinear, GraN-DAG) cannot run at $d \geq 100$ and SCORE did not finish within our budget.

d	Method	AUROC	AUPRC	seeds
50	DAGMA-linear	0.648±0.046	0.138	3
50	GOLEM	0.807±0.043	0.188	3
50	NOTEARS-MLP	0.530±0.050	0.023	3
50	NOTEARS-linear	0.801±0.018	0.355	3
50	SPADE (λ_g=0.01)	0.815±0.039	0.438	3
50	SPADE-scaled	0.961±0.011	0.774	3
100	DAGMA-linear	0.655±0.031	0.116	3
100	NOTEARS-linear	0.578±0.031	0.069	3
100	SPADE (λ_g=0.01)	0.678±0.026	0.043	3
100	SPADE-scaled	0.982±0.009	0.830	3
200	DAGMA-linear	0.674±nan	0.155	1
200	NOTEARS-linear	0.471±nan	0.007	1
200	SPADE (λ_g=0.01)	0.579±nan	0.006	1
200	SPADE-scaled	0.979±nan	0.766	1

Table 4: Shape fidelity on fresh seeds 400–419. MLP-edge: same bottleneck, penalty and budget with $1 \to 3 \to 1$ $\tanh$ edges. SPADE-2stage: structure from the plain SPADE fit, curves from the smoothed refit.

		found edges (all)		found, strongly non-linear	
d	Method	R^2	class agr.	R^2	(n)
6	MLP-edge	0.92	0.91	0.23	(4)
6	SPADE	0.91	0.25	0.71	(16)
6	SPADE-2stage	0.96	0.93	0.85	(16)
10	MLP-edge	0.95	0.94	0.24	(3)
10	SPADE	0.91	0.12	0.55	(15)
10	SPADE-2stage	0.96	0.94	0.63	(15)

3.2 Scale: $d = 50$ to 200

Table 3 shows the fixed penalty collapsing as d grows, while the width-aware rule with screening holds AUROC near 0.96–0.98. The comparison at $d \geq 100$ is only against linear methods, which are mis-specified for this data; it shows SPADE-scaled works at this size, not that it beats the best non-linear learner. Run times are not reported because jobs shared a four-core machine.

3.3 Interpretability, measured

For every true edge that a method ranks within its top-$|E|$ scored edges, we compare the learned curve with the true mechanism: R^2 of the best affine map, and agreement of the qualitative class (increasing, decreasing, non-monotone). "Strongly non-linear" edges are those a straight line explains less than half of (sin and quadratic). Table 4 has two messages. First, raw SPADE curves recover the right values ($R^2 \approx 0.91$) but the wrong qualitative class on most edges, because of wiggles; the smoothed display refit repairs this (0.93–0.94) and reaches the best R^2 (0.96). Second, on strongly non-linear edges, SPADE finds many more of them than the matched-capacity MLP edges and recovers their shape far better. The "found" rule uses the true number of edges, so it favours no method but is an oracle count. A metric computed in an early run

Table 5: Sachs, 10 subsamples of 2,000 cells (chance AUPRC $= 0.164$). SPADE-AB and plain SPADE use the settings fixed on synthetic data.

Method	AUROC	AUPRC	top-18 hits	SHD
SPADE-AB	0.664±0.043	0.319	6.0	24.0
SPADE	0.601±0.048	0.289	5.8	24.4
NOTEARS-linear	0.566±0.015	0.239	5.0	26.0
GOLEM	0.528±0.037	0.195	3.6	28.8
DAGMA-nonlinear	0.486±0.117	0.238	3.8	28.4
GraN-DAG	0.484±0.002	0.164	2.0	32.0
NoGAM	0.465±0.040	0.157	6.2	50.0
SCORE	0.461±0.037	0.156	6.0	49.8
DAGMA-linear	0.432±0.035	0.167	3.0	30.0
NOTEARS-MLP	0.297±0.036	0.117	0.1	35.8

was wrong because of an indexing error (the curve was read from the wrong input column); we fixed it and re-ran before any result was used.

3.4 A real benchmark with ground truth: Sachs

The Sachs data are 7,466 single-cell measurements of 11 phospho-proteins with an 18-edge consensus graph. We use 10 random subsamples of 2,000 cells after a log transform; no method is tuned on these data. SPADE-AB is best (Table 5); it beats every other method in 8 or more of 10 resamples ($p \leq 0.010$). Absolute accuracy is low: it recovers about 6 of the 18 edges among its top 18. Sachs is difficult for all methods, and this result shows relative rather than good performance.

4 Limitations

(1) Gains from SPADE-AB are modest. (2) At $d \geq 100$ only linear baselines were run. (3) The parent screening step can discard true parents, so recall is bounded by the screen. (4) The interpretability metric uses synthetic mechanisms and an oracle edge count; it is not a human study. (5) Sachs is a single real data set. (6) On the financial and weather series we tried, no learned one-step forecaster beat a naive last-value prediction, SPADE variants included, so we make no forecasting claim.

5 Conclusion

A width-aware penalty with parent screening makes SPADE usable at $d = 200$; adaptive, averaged sparsity gives a small but consistent accuracy gain; and a smoothed display refit makes the learned curves match the true mechanism's class without altering the recovered graph. The next step is a comparison with non-linear learners at $d \geq 100$ and a human study of whether the curves change decisions.

References

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